$$X_L = 2\pi fL = 2\pi \times 50 \times \frac{2}{\pi} = 200\,\Omega$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times \frac{10^{-4}}{\pi}} = 100\,\Omega$$

(i) Impedance, $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$$= \sqrt{100^2 + (200 - 100)^2} \quad = 141.4\ \Omega$$

(ii) Peak value of current,

$$I_m = \frac{V_m}{Z} = \frac{\sqrt{2}\,V_{RMS}}{Z}$$

$$= \frac{\sqrt{2} \times 220}{141.4} \quad = 2.2\ \text{A}$$

(iii) Power factor of the circuit,

$$\cos\phi = \frac{R}{Z} = \frac{100}{141.4} = 0.707$$

(iv) Power factor at resonance,

$$\cos\phi = \frac{R}{Z} = \frac{R}{R} = 1$$

4.9 OSCILLATION IN LC CIRCUITS

4.9.1 Energy conversion during LC oscillations

We have learnt that energy can be stored in both inductors and capacitors (Refer sections 1.8.2 and 4.3.2). In inductors, the energy is stored in the form of magnetic field while in capacitors, it is stored as the electric field.

Whenever energy is given to a circuit containing a pure inductor of inductance L and a capacitor of capacitance C, the energy oscillates back and forth between the magnetic field of the inductor and the electric field of the capacitor. Thus the electrical oscillations of definite frequency are generated. These oscillations are called ***LC* oscillations**.

Generation of LC oscillations

Let us assume that the capacitor is fully charged with maximum charge Q_m at the initial stage. So that the energy stored in the capacitor is maximum and is given by $U_E = {Q_m^2}/{2C}$. As there is no current in the inductor, the energy stored in it is zero i.e., $U_B = 0$. Therefore, the total energy is wholly electrical. This is shown in Figure 4.52(a).

The capacitor now begins to discharge through the inductor that establishes current i in clockwise direction. This current produces a magnetic field around the inductor and the energy stored in the inductor is given by $U_B = {Li^2}/{2}$. As the charge in the capacitor decreases, the energy stored in it also decreases and is given by $U_E = {q^2}/{2C}$. Thus there is a transfer of some part of energy from the capacitor to the inductor. At that instant, the total energy is the sum of electrical and magnetic energies (Figure 4.52(b)).

When the charges in the capacitor are exhausted, its energy becomes zero i.e., $U_E = 0$. The energy is fully transferred to the magnetic field of the inductor and its energy is maximum. This maximum energy is given by $U_B = {LI_m^2}/{2}$ where I_m is the

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